

Corpus Linguistic Toolkit for Analysing (Brazilian) Portuguese

Universität Konstanz  
Corpus Linguistics with Python  
Summer Semester 2026  
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Speech and Language Processing  
August 31st, 2026

## **1 Project overview**

This project was created as a final assignment for the course Corpus Linguistics with Python, taught by Prof. Dr. Kengatharaiyer Sarveswaran at the University of Konstanz. It consists of a corpus analysis toolkit for (Brazilian) Portuguese. The functionalities of the program include corpus collection and organization, pre-processing of the data, linguistic annotation, data analysis, corpus analysis, corpus search and N-gram analysis.

After corpus collection (which will be better explained in the next section), corpus organization consisted of categorizing and storing basic information of the different texts in the corpus (title, source, author, year, genre, copyright/license ) on a Json file which can be accessed by the user. Pre-processing included sentence segmentation, corpus tokenization, normalization of the text string (lowercase) and removal of punctuation, creating a list of clan words, as well as stop word removal, creating a list of content words. Considering how the texts all had the same source, genre and licence/copyright, the extraction/filtering function was not included.

The linguistic annotation function features a Part-of-Speech tagger, downloaded from the SpaCy module ("pt\_core\_news\_sm"), which tags every token in the corpus with a grammatical form (and will later on be used for corpus search). The data analysis section features word frequency counters as well as character frequency counters and POS frequency counters. The corpus analysis section includes a Token-Type Ratio (TTR) indicator and a Keyword in Context search. For corpus search, the possibilities are keyword search, pattern search and POS-tag search. Lastly, it is possible to display the top 10 most frequent unigrams, bigrams and trigrams in the selected corpus.

## **2 Data collection approach and corpus description**

The corpus was collected through web crawling/scraping program built into the project. It features 21 research articles on the topics of Biology, Sociology and Literature and was collected from the Scientific Electronic Library Online (SciELO), which is an online library of free access. This source was chosen due to the free access and use policy as well as the high volume of good-quality content the genre scientific article offers. One problem with this genre, however, is that it often includes at least one section (usually a paragraph) written in a foreign language (mostly English and Spanish), causing the corpus to not be pure.

## **3 System architecture/design decisions**

The way the program was designed was informed by the functionalities it presents. First corpus collection, then pre-processing and so on.

## **4 Usage instructions**

The program is better navigated as a Jupyter Notebook. Each functionality is marked with a comment for easier navigation and some offer the possibility of interaction, such as corpus search.

## **5 Example commands**

The following command asks for input, where the user can type a Part-of-Speech tag and receive as output all of the tokens marked with that specific tag.

### # Annotation-based search

```
tokens_pos = []
POS_tag = input("Please enter a POS for search: \n")
matching_tokens = [token.text for token in doc if token.pos_ == POS_tag]
print(matching_tokens)
Please enter a POS for search:
NOUN
```

The following command allows the user to type a word and receive the context surrounding that word:

```
# Concordance generation/KWIC (Key Word in Context) analysis

kc = input("Please enter a keyword for context search: \n")
sample_nltk_text = Text(final_tokens)
sample_nltk_text.concordance(keyword_context)
```

## 6 Example outputs

Total number of tokens:

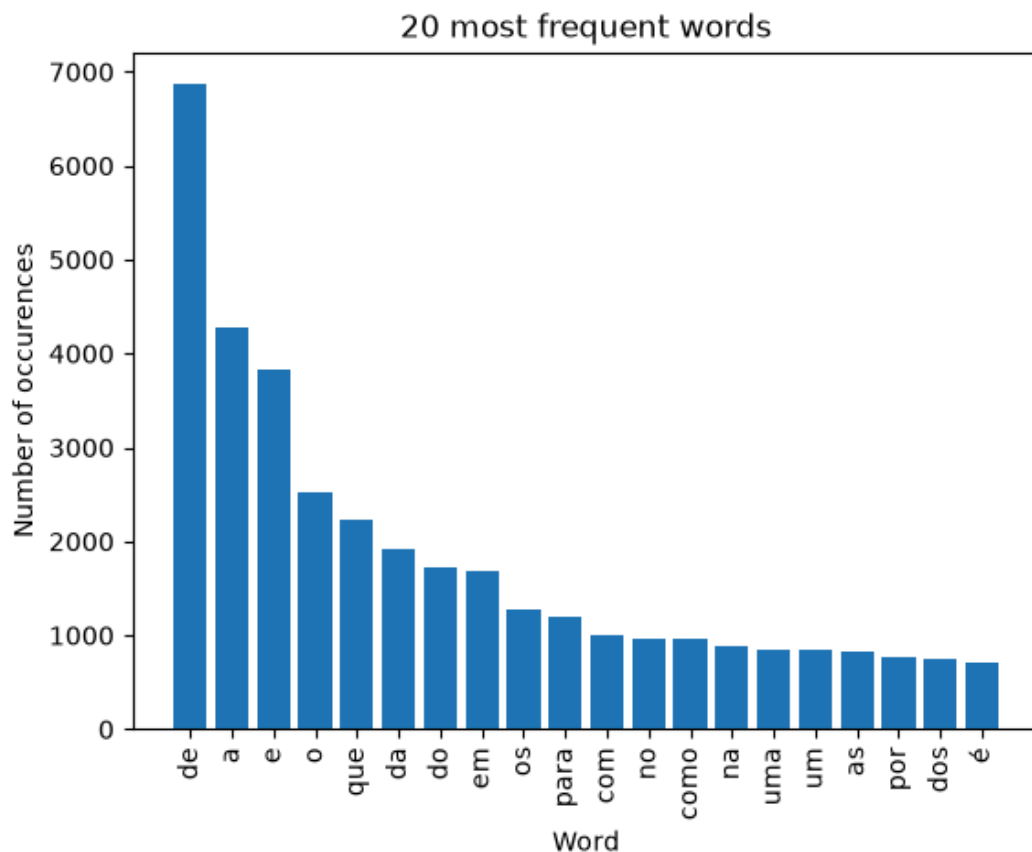
Number of tokens: 119954

20 most common words:

The 20 most common words in the corpus before stop word removal are:

```
[('de', 4778), ('a', 4092), ('e', 3386), ('o', 2874), ('que', 2072), ('do', 1957), ('da', 1681), ('em', 1595), ('os', 1066), ('no', 972), ('com', 857), ('por', 848), ('um', 847), ('para', 842), ('é', 827), ('uma', 823), ('como', 754), ('na', 742), ('as', 720), ('dos', 639)]
```

Bar chart of the 20 most frequent words in the corpus



## 7 Challenges faced

There were three big challenges. The first one involved the correct selection of source for the corpus. Some websites did not allow the copying of their data (which indicates the possibility of copyright violation). Some others allowed it but the structural system was very busy, with multiple elements interpolating one another and links which confused the scraping system (such as Wikipedia). The most important thing, however, was coming across a web page with substantial information (i.e. words) but that allowed complete use and manipulation of the content; most of them only allowed partial use.

The second challenge was collocation analysis using measures such as Mutual Information (MI), t-score, Dice coefficient, log-likelihood, which could not be completed and therefore included in the final project due to the high level of complexity.

Lastly, the debugging. At almost every step in the construction of this project there was a different error that needed to be fixed for the functionality to perform. They usually took some time to fix , and any alternation in one part of the program could affect another, causing it to malfunction.

## References

- Sweigart, A.. (2015). Automate the boring stuff with Python: practical programming for total beginners. No Starch Press.
- SciELO. (n.d.). SciELO Brazil: Scientific Electronic Library Online. <https://www.scielo.br/>.